Deep Learning for Natural Language Processing

Today's goal • 3/3 items

Module 1
Feedforward Neural Nets and Recurrent Neural Networks

Introduction to the Course
- Course Updates and Accessibility Support — Reading • 1 min
- Earn Academic Credit for Your Work! — Reading • 10 min
- Course Support — Reading • 10 min
- Assessment Expectations — Reading • 5 min
- AI Citation and Acknowledgement — Reading • 10 min

Feedforward Networks
- The Perceptron — Video • 6 min
- The XOR Problem — Video • 4 min
- Feedforward Networks — Video • 7 min
- Feedforward Networks in NLP — Video • 8 min
- Mathematics of Feedforward Networks — Reading • 25 min
- Sentence Embeddings with Feedforward Networks — Video • 5 min
- Feedforward Networks — Practice Assignment • Grade: 100%

Recurrent Neural Networks
- Sentence Embeddings with Recurrent Neural Networks — Video • 4 min
- Sequence Labelling with Recurrent Neural Networks — Video • 4 min
- Mathematics of Recurrent Neural Networks — Reading • 25 min
- LSTMs and GRUs — Video • 6 min
- Bidirectional Recurrent Neural Networks — Video • 3 min
- Hierarchical Recurrent Neural Networks — Video • 5 min
- Recurrent Neural Networks in Python — Video • 8 min
- Recurrent Neural Networks — Practice Assignment • Grade: 100%

Implementing and Training Neural Networks
- Loss Functions — Video • 6 min
- The Intuition behind Gradient Descent — Video • 5 min
- Stochastic Gradient Descent — Video • 6 min
- Stochastic Gradient Descent — Reading • 25 min
- Stochastic Gradient Descent in Python — Video • 6 min
- Stochastic Gradient Descent — Lab • 30 min
- Training Neural Networks — Practice Assignment • Grade: 100%

Review and Exercises
- AI Policy Quiz — Graded Assignment • Grade: 100%
- Feedforward Networks, Recurrent Neural Networks, and How to Train Them — Graded Assignment • Grade: 100%
- **Recurrent Neural Networks for Sequence Labelling**

Programming assignment

Recurrent Neural Networks for Sequence Labelling

Launch lab



What to expect
- Due Nov 9, 11:59 PM IST
- Unlimited attempts

You passed!
To pass you need at least 80/100 points.

100/100

Instructions My submissions Discussions

This lab introduces Recurrent Neural Networks (RNNs) and their application to sequence labeling tasks, specifically Part-of-Speech (POS) tagging using the Brown corpus.

Key Concepts:

1. Recurrent Neural Networks (RNNs):
   - Neural networks designed to work with sequential data
   - Process input sequences one element at a time while maintaining a hidden state
   - Can handle variable-length sequences
   - Well-suited for tasks like language processing, time series analysis, and sequence labeling
2. Long Short-Term Memory (LSTM):
   - A sophisticated type of RNN that addresses the vanishing gradient problem
   - Contains specialized gates (input, forget, output) to control information flow
   - Better at capturing long-term dependencies in sequences
   - More stable training compared to vanilla RNNs
3. Sequence Labeling:
   - Task of assigning a categorical label to each element in a sequence
   - Examples include POS tagging, named entity recognition, and chunking
   - Each input token maps to exactly one output label
4. Part-of-Speech (POS) Tagging:
   - Linguistic task of marking words with their grammatical categories
   - Examples: noun, verb, adjective, determiner, etc.
   - Context-dependent: same word can have different POS tags based on usage
   - Essential for many downstream NLP tasks
5. Key Components in Implementation:
   - Word Embeddings: Dense vector representations of words
   - LSTM Layer: Processes the sequence and captures contextual information
   - Output Layer: Maps LSTM outputs to tag probabilities
   - Cross-Entropy Loss: Standard loss function for classification tasks

In this lab, we'll implement a complete sequence labeling system using PyTorch, demonstrating how to:

- Process and prepare sequential data
- Build an RNN-based model architecture
- Train the model using backpropagation through time
- Evaluate the model's performance on a held-out test set

The Brown corpus serves as our dataset, providing pre-tagged sentences that we'll use for training and testing our model.

Report an issue

Go to next item →